

■ FootballPredict Platform

Full System Architecture, Technical Specification & Engineering Documentation

System Release:	v2.4 Enterprise Production
Production Frontend:	https://footballprediction-lovat.vercel.app
Production Backend API:	https://football-prediction-api-mmet.onrender.com
AdSense Publisher ID:	pub-3089881788835574
Admin Account:	Wes@254 (davidwesonga776@gmail.com)
Classification:	Technical & Architecture Blueprint

1. Executive Summary & Platform Overview

FootballPredict is a full-stack, enterprise-grade football intelligence and multi-market prediction ecosystem. The platform provides global football supporters with real-time match fixture schedules, live scores, head-to-head tactical metrics, expected goals (xG) projections, and statistical probabilities across 4 betting markets (1X2 Match Outcome, Over/Under 2.5 Goals, Both Teams to Score, and Double Chance).

Additionally, the platform incorporates an AI conversational assistant (**Coach AI**), a competitive community leaderboard with automated scoring rules, transactional email notifications (welcome & daily match digests), a secure administrative telemetry dashboard, and integrated **Google AdSense** monetization with GDPR-compliant privacy management.

2. System Architecture & Topology

The system employs a decoupled, cloud-native client-server architecture with high availability and automated CI/CD pipelines.

Layer	Technology Stack	Key Responsibilities
Frontend Client	Next.js 16 (App Router), React 19, TypeScript, Turbopack, Tailwind CSS	Server & Client Components, responsive UI, date grouping, universal search, AdSense units, Coach AI modal, theme provider.
Backend API	FastAPI, Python 3.14, Pydantic v2, Uvicorn, BackgroundTasks	RESTful endpoints, JWT security, Poisson ML engine, email dispatchers, rate limiters, admin telemetry.

Persistence Layer	PostgreSQL / SQLite, SQLAlchemy ORM	Users, Matches, Leagues, Teams, Predictions, and Leaderboard aggregation schemas.
External Integrations	football-data.org, The Odds API, Google AdSense, SMTP Server	Live match schedules, bookmaker odds feeds, ad serving, and transactional email deliverability.

3. Mathematical Modeling & ML Prediction Engine

Match probabilities are computed using a bivariate **Double Poisson Goal Model** combined with dynamic team attacking/defending ratings, recent 5-game form factors, and home-field advantage weighting.

3.1 Poisson Goal Expectancy Equation

For a home team with expected goals λ_{home} and an away team with expected goals μ_{away} , the joint probability of an exact scoreline (x, y) is defined as:

$$P(\text{Home} = x, \text{Away} = y) = [(\lambda^x e^{-\lambda}) / x!] \times [(\mu^y e^{-\mu}) / y!]$$

3.2 Multi-Market Derivations

- **1X2 Match Winner:** Evaluated across a matrix of scorelines up to 7-7 (Home Win: $x > y$, Draw: $x = y$, Away Win: $x < y$).
- **Over / Under 2.5 Goals:** Probability that total match goals $(x + y) \geq 3$ vs $(x + y) \leq 2$.
- **Both Teams to Score (BTTS):** Probability that both $x \geq 1$ and $y \geq 1$.
- **Double Chance:** $P(1X) = P(\text{Home}) + P(\text{Draw})$, $P(X2) = P(\text{Away}) + P(\text{Draw})$, $P(12) = P(\text{Home}) + P(\text{Away})$.

4. Database Architecture & Data Models

Model	Primary Fields	Relationships & Indexes
User	id, username, email, hashed_password, is_admin, is_active, total_points, accuracy	One-to-Many with Prediction. Unique indexes on username and email (case-insensitive).
League	id, code, name, country, flag, is_active	One-to-Many with Match and Team. Key codes: PL, CL, PD, BL1, SA, FL1, ELC, DED.
Team	id, external_id, name, short_name, tla, crest_url, form	Associated with home and away fixtures. Attack and defense ratings.
Match	id, external_id, utc_date, status, home_score, away_score, odds_*, prob_*	Foreign keys to League, Home Team, and Away Team. Indexed by utc_date & status.
Prediction	id, user_id, match_id, predicted_home, predicted_away, points_awarded, is_scored	Composite unique constraint on (user_id, match_id). Foreign keys to User and Match.

5. API Directory & Integration Specifications

Route	Method	Access	Functionality
/api/auth/register	POST	Public	Registers user, checks uniqueness, sends welcome email.
/api/auth/login	POST	Public	Authenticates credentials, issues JWT bearer token.
/api/matches	GET	Public	Fetches matches filtered by status, date, and league.
/api/predictions	POST	User	Submits exact scoreline prediction before kickoff.
/api/leaderboard	GET	Public	Returns ranked leaderboard entries by points and accuracy.
/api/chat	POST	Public	Processes Coach AI conversational football reasoning.
/api/admin/stats	GET	Admin	Returns telemetry counts for matches, users, and SMTP.
/api/admin/sync	POST	Admin	Triggers asynchronous full competition data sync.
/api/admin/score	POST	Admin	Calculates leaderboard points for finished matches.

6. Security Architecture & Administration Controls

The system enforces a zero-trust, defense-in-depth security model across authentication, session management, and administrative privileges.

6.1 Authentication & Password Security

- **Bcrypt Password Hashing:** Passwords are cryptographically salted and hashed using PassLib Bcrypt.
- **JWT Tokenization:** HS256 algorithm generates digitally signed JSON Web Tokens for stateless API calls.
- **Case-Insensitive Duplicate Validation:** Prevents spoofing or dual registration by enforcing lowercase email & username validation.

6.2 Admin Inactivity Auto-Logout Timer

To protect elevated controls, the Admin Dashboard implements a **5-minute (300-second) inactivity detector**. Continuous listener monitors user interaction events (mousemove, keydown, mousedown, touchstart, scroll, click). If idle for 300 seconds, the admin session is revoked, tokens are purged, and the admin is safely logged out.

7. Monetization Engine & Google AdSense Setup

The monetization architecture incorporates a dual-mode strategy designed for maximum CPM yields and compliance.

7.1 Ad Unit Topology & Layout Locations

1. **Hero Top Banner:** High-visibility placement above the daily match feed.
2. **In-Feed Match Break Banner:** Embedded between date clusters in the fixture schedule.
3. **VIP Tool & Forecaster Banner:** Embedded within secondary match blocks.
4. **Merchandise & Gear Banner:** Positioned below the match listings.
5. **Leaderboard & Rewards Footer:** Placed on the competitive rankings page.

7.2 Verification & Compliance Specification

- **Publisher ID:** pub-3089881788835574 (Client ID: ca-pub-3089881788835574)
- **Authorized Digital Sellers:** Hosted at /ads.txt with DIRECT reseller authorization.
- **GDPR & European Privacy:** Google Certified CMP integration for EEA, UK, and Swiss user consent management.

8. Gamification & Prediction Scoring Matrix

Prediction Outcome	Points Awarded	Example Scenario
Exact Scoreline Hit	+3 Points	User predicted 2 - 1; Final Result: 2 - 1.
Correct Match Winner / Draw	+1 Point	User predicted 3 - 0 (Home Win); Final Result: 1 - 0 (Home Win).
Incorrect Result	0 Points	User predicted 2 - 0 (Home Win); Final Result: 1 - 2 (Away Win).

9. Verification & Sign-off

This document confirms that all components of the **FootballPredict Platform** have been engineered, tested, and validated against production benchmarks. Continuous deployments are monitored via Vercel Edge and Render Cloud Infrastructure.